24. Elimination Theory and General Ideal Theory

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In what follows the question is the placement of elimination theory – in the arithmetical form that I gave to Hentzelt's presentation[1] and that may be described as an ideal theory in the polynomial domain – within general ideal theory,[2] and, at the same time, a new foundation of the parts referring to zeroes. The basic concepts of both theories are briefly assembled in § 1, as are those of field theory,[3] so that I can refer to them here.

The classification and the new foundation group themselves around four questions: the arithmetical formulation of the concept of dimension; the theory of zeroes; the connection between the decomposition theorems for norms and elementary divisors and those of general ideal theory; the characterization of prime ideals and primary ideals by norm and elementary-divisor form.

The arithmetical formulation of the *concept of dimension* (§ 4) is given by chains of prime ideals, the arithmetical equivalent of the fact that on surfaces there are curves, and on curves there are points. This arithmetical formulation, which is proved identical with the parameter definition of elimination theory, can be transferred with a slight modification to arbitrary ring domains and, together with the transfer of certain parts of the other questions, gives there an exact insight into the structure of ideals, as I shall show elsewhere.

In H.-N. the question of *zeroes* is, as usual, attacked directly for the given ideal, namely by going back to successive elimination, whereby the character of verification is not entirely avoided. In contrast stands the route always taken for polynomials in one variable: one represents the given polynomial as a product of powers of prime functions (primary functions) and constructs, for each individual prime function, the field of zeroes as a field isomorphic to the residue-class field. The degree of the prime function gives the degree of this field; the degree of the primary function, however, whose zeroes coincide with those of the prime function, gives the degree of the ring of residue classes modulo this primary function. If, for each prime function, one passes to the Galois field, one obtains the decomposition into linear factors; and with this, for the originally given polynomial, the question of zeroes, decomposition into linear factors, and multiplicity is settled. Quite correspondingly, here (§ 3) the system of residue classes modulo a prime ideal is extended, by forming quotients, to the residue-class field, and a field of zeroes is constructed that is isomorphic to it. This field of zeroes contains a subfield generated by adjoining indeterminates – say $x_{i+1}, \ldots, x_n$; the degree of the norm gives the degree with respect to this subfield; at the same time, by passage to the Galois field, the decomposition of the elementary-divisor form and hence of the norm into linear factors is obtained. For the corresponding primary ideal, however, the zeroes are the same; the decomposition is also given along with it, since the norm becomes a power of the elementary-divisor form of the prime ideal (§ 2); the degree of the norm gives the degree of the ring of residue classes modulo the primary ideal with respect to the subfield derived from the indeterminates.

Thus the question of zeroes of an arbitrary ideal is reduced to the connection between the decomposition theorems for norms and elementary divisors and those of general ideal theory (§ 5). The fact that the zeroes of the ideal are composed from those of its individual associated prime ideals corresponds to the decomposition of the norm into greatest primary factors, which become equal to powers of the elementary-divisor form of the individual associated prime ideals;[4] thereby the decomposition into linear factors is accomplished for the norm in general. Multiplicity, however, receives its interpretation in H.-N. by means of the fundamental ideals and their components; the degree of a greatest primary factor of the norm is made equal to the number of linearly independent residue classes – after adjoining $x_{i+1}, \ldots, x_n$ – of the fundamental ideal of $(i-1)$-st stage modulo a component of the fundamental ideal of i-th stage. Now the fundamental ideal of $(i-1)$-st stage is proved identical with the isolated component of the decomposition that is uniquely defined by all prime ideals of dimension higher than the $(n-i)$-th; and a component of the fundamental ideal of i-th stage is proved identical with the isolated component of the decomposition defined by the prime ideal corresponding to the factor of the norm and by all prime ideals of higher dimension. The degree of the corresponding factor of the norm becomes – after adjoining $x_{i+1}, \ldots, x_n$ – equal to the degree of that subring of the residue classes of this component that consists only of classes of the fundamental ideal of $(i-1)$-st stage.

The characterization of the prime ideals and proper primary ideals (§ 6) results as a direct consequence of the consideration of the residue-class fields. If the original coefficient domain is a perfect field, then prime

[1]Kurt Hentzelt, Zur Theorie der Polynomideale und Resultanten. Edited by E. Noether. Math. Ann. 88 (1922), pp. 53–79; cited H.-N.

[2]E. Noether, Idealtheorie in Ringbereichen, Math. Ann. 83 (1921), pp. 24–66; cited Ideal Theory.

[3]E. Steinitz, Algebraische Theorie der Körper, J. f. M. 137 (1910), pp. 167–309; cited Steinitz.

[4]This parallelism between elimination theory and general ideal theory is not fulfilled in Kronecker's elimination theory; compare H.-N., note *).

functions correspond to the prime ideals as norm and elementary-divisor form, proper primary functions to the proper primary ideals, and conversely. For an imperfect field as coefficient domain only conditions can be given that are either necessary or sufficient; as examples show, the norm of a prime ideal can become properly primary, and the elementary-divisor form of a proper primary ideal can become a prime function. More precisely, in characteristic p the norm of a prime ideal becomes the p^f-th power ($f > 0$) of its elementary-divisor form; whereas, for proper primary ideals whose elementary-divisor form is a prime function, the norm can become an arbitrary power, as examples again show. Finally, absolute prime ideals are considered (§ 7), that is, those that remain prime ideals when the coefficient domain is extended to an algebraically closed field. For absolute prime ideals with coefficients from a (finite) algebraic number field, the characterization leads to the transfer of a theorem first stated by A. Ostrowski for absolute prime functions: the property of being a prime ideal is preserved modulo every prime ideal of the number field, with at most finitely many exceptions.

Let the analogy of the last question with ideal theory in algebraic number fields also be pointed out. As the elementary-divisor form of such an ideal one has to regard the least integral rational number divisible by the ideal (note 10), which for a prime ideal therefore always becomes a prime number, whereas conversely an ideal becomes a prime ideal as soon as its norm is a prime number; the proofs also run quite in parallel. The examples mentioned above thus show that, over an imperfect field, the analogue of prime ideals of higher degree and ramification ideals occurs, whereas over a perfect field there are only prime ideals of first degree and no ramification ideals.

§ 1. Basic Concepts of Elimination Theory, General Ideal Theory, and Field Theory

1. The domain. Let $\mathfrak{S}$ denote the integral domain of all polynomials in $y_1, \dots, y_n$ with coefficients from an abstractly defined field P. Let the y be subjected to a transformation with indeterminates $u_{\mu\nu}$ as coefficients:[5]

$$y_1 = u_{11}x_1 + \cdots + u_{1n}x_n, \quad \dots, \quad y_n = u_{n1}x_1 + \cdots + u_{nn}x_n, \tag{1}$$

with inverse

$$x_1 = t_{11}y_1 + \cdots + t_{n1}y_n, \quad \dots, \quad x_n = t_{1n}y_1 + \cdots + t_{nn}y_n. \tag{1'}$$

The $u_{\mu\nu}$, or – what is equivalent because of mutual rational expressibility – the $t_{\mu\nu}$, are adjoined to P, so that the coefficient domain $P(u) = P(t)$ arises; let $\mathfrak{S}'$ denote the integral domain of all polynomials in $x_1, \dots, x_n$ with coefficients from $P(u)$. The domains $\mathfrak{S}$ and $\mathfrak{S}'$ underlie elimination theory; ideals $\mathfrak{a}, \mathfrak{b}, \dots$ in $\mathfrak{S}$ and $\mathfrak{a}', \mathfrak{b}', \dots$ in $\mathfrak{S}'$ are considered. An ideal in an arbitrary ring is here defined in the usual way by the requirement that, together with α and β, it also contain $\alpha - \beta$, and, together with α, also $\lambda\alpha$, where λ is an arbitrary ring element; $\mathfrak{o}$ will always denote the unit ideal consisting of all elements.

2. Transformed ideals. An ideal $\mathfrak{m}$ in $\mathfrak{S}'$ is called transformed if it arises from an ideal $\overline{\mathfrak{m}}$ in $\mathfrak{S}$ by means of (1) after adjoining the $u_{\mu\nu}$, or the $t_{\mu\nu}$. Thus $\mathfrak{m}$ consists, when $\bar{f}(y)$ or $\bar{g}(y)$ runs through all polynomials in $\overline{\mathfrak{m}}$, of all linear combinations

$$f(x) = \sum U_i(u)\bar{f}_i(y) = \sum U_i(u) f_i(x)$$

or also

$$g(x) = \sum T_i(t)\bar{g}_i(y) = \sum T_i(t) g_i(x).$$

In particular all $f_i(x) = \bar{f}_i(y)$ are contained, or, what amounts to the same thing, all $g_i(x) = \bar{g}_i(y)$. Since $f(x)$ respectively $g(x)$ are fixed only up to quantities from $P(u) = P(t)$, the U_i, T_i may, without loss of generality, be assumed to be power products of the u, respectively the t. The polynomials $f(x), g(x)$ again pass into polynomials of $\mathfrak{m}$ if U, T are replaced by arbitrary other power products, in particular by interchanging the columns of (1), respectively the rows of (1'); hence $\mathfrak{m}$ contains, together with any polynomial, all those arising from it by interchanging the x, provided only that the corresponding interchanges are carried out in the coefficients depending on u, respectively t. All ideals occurring in what follows are to be assumed transformed unless the contrary is expressly stated. Non-transformed ideals pass, by composing (1) with

$$x_i = v_{i1}z_1 + \cdots + v_{in}z_n,$$

into transformed ideals in the polynomial domain of the z with coefficients from $P(u, v)$, whereas for transformed ideals this composition amounts merely to replacing the $u_{\mu\nu}$ by bilinear combinations of the u, v.

[5] The transformation is, with a view to § 3 and so on, somewhat more general than in H.-N.; all results there are thereby retained all the more.

3. Notation. By $f^{(i)}, g^{(i)}, \ldots, a^{(i)}, b^{(i)}, \ldots$ we shall throughout understand polynomials in $\mathfrak{S}'$ that are free of $x_i, \ldots, x_{i-1}$.

4. Fundamental ideals. To every ideal $\mathfrak{m}$ there are assigned n fundamental ideals $\mathfrak{g}_0, \mathfrak{g}_1, \ldots, \mathfrak{g}_{n-1}$ by the following stipulation: the fundamental ideal of $(i-1)$-st stage, $\mathfrak{g}_{i-1}$, contains all and only those polynomials $G(x)$ for which there exists a polynomial $b^{(i)} \neq 0$ – in general varying with $G(x)$ – such that $b^{(i)}G(x) \equiv 0 \pmod{\mathfrak{m}}$. From the existence of the ideal basis follows also the existence of at least one $B^{(i)}$, fixed for all $G(x)$, so that

$$B^{(i)}\mathfrak{g}_{i-1} \equiv 0 \pmod{\mathfrak{m}}, \qquad B^{(i)} \neq 0. \tag{2}$$

The fundamental ideals of transformed ideals themselves become transformed ideals (H.-N., Theorem VI). The fundamental ideal of i-th stage $\mathfrak{g}_i$ is also defined as the ideal into which $\mathfrak{m}$ passes when $x_{i+1}, \ldots, x_n$ are adjoined to $P(u)$, provided one restricts oneself – which then means no loss of generality – to polynomials integral in $x_{i+1}, \ldots, x_n$; $\mathfrak{g}_0$ is always equal to the unit ideal $\mathfrak{o}$.

5. Module representation of ideals, elementary divisors and individual norms. If every polynomial $f(x)$ is regarded as a linear form in the power products ξ of $x_1, \ldots, x_{i-1}$, with polynomials $a^{(i)}$ as coefficients, then the ideal $\mathfrak{m}$ passes into a module $\mathfrak{M}_{i-1}$ of linear forms with respect to the domain of the $a^{(i)}$; that is, $\mathfrak{M}_{i-1}$ contains, together with any two linear forms, also their difference, and, together with a linear form $l(\xi)$, also $a^{(i)}l(\xi)$ for arbitrary $a^{(i)}$. As fundamental module of such a module $\mathfrak{M}$ one denotes the totality of linear forms $g(\xi)$ for which $b^{(i)}g(\xi) \equiv 0 \pmod{\mathfrak{M}}$, $b^{(i)} \neq 0$; hence the fundamental ideal $\mathfrak{g}_{i-1}$ passes, under the module representation, into the fundamental module $\mathfrak{G}_{i-1}$ of $\mathfrak{M}_{i-1}$. $\mathfrak{M}_{i-1}$ and $\mathfrak{G}_{i-1}$ depend on infinitely many indeterminates ξ, in such a way that in each individual linear form only finitely many of these indeterminates occur. $\mathfrak{M}_{i-1}$ has the characteristic property that, after adjoining $x_{i+1}, \ldots, x_n$ to $P(u)$, it has only finitely many elementary divisors different from unity; that is, after this adjunction $\mathfrak{G}_{i-1}$ and $\mathfrak{M}_{i-1}$ admit a basis representation – with ζ denoting new indeterminates, connected with the ξ by invertible linear transformations integral in x_i, in such a way that in $\zeta_i = r_i(\xi)$ only finitely many of these indeterminates occur at a time –

$$\begin{aligned}
\mathfrak{G}_{i-1} &= (\zeta_0, \zeta_1, \ldots, \zeta_\sigma, \zeta_{\sigma+1}, \ldots, \zeta_{\sigma+\nu}, \ldots), \\
\mathfrak{M}_{i-1} &= (E_0^{(i)}\zeta_0, E_1^{(i)}\zeta_1, \ldots, E_\sigma^{(i)}\zeta_\sigma, \zeta_{\sigma+1}, \ldots, \zeta_{\sigma+\nu}, \ldots),
\end{aligned} \tag{3}$$

where each $E_\mu^{(i)}$ divides the preceding one (H.-N. (24) and (28)).

The highest elementary divisor $E^{(i)}$ is also defined as the greatest common divisor – in the polynomial sense – of all $B^{(i)}$ occurring in (2); it can be assumed integral and primitive in $x_{i+1}, \ldots, x_n$, and is then regular in x_i, that is, $E^{(i)}$ contains the term x_i^λ with non-zero coefficient, if it is of degree λ in the x.

The product $R^{(i)}$ of all elementary divisors occurring in (3) is called the norm of $\mathfrak{G}_{i-1}$ with respect to $\mathfrak{M}_{i-1}$, or also the individual norm of the ideal $\mathfrak{m}$; in symbols, taking account of the fact that $\mathfrak{m}$ passes into $\mathfrak{g}_i$ when $x_{i+1}, \ldots, x_n$ are adjoined:

$$R^{(i)} = E_0^{(i)}E_1^{(i)} \cdots E_\sigma^{(i)} = N(\mathfrak{G}_{i-1} \mid \mathfrak{M}_{i-1}) = N(\mathfrak{g}_{i-1} \mid \mathfrak{g}_i). \tag{1}$$

As (3) shows, after adjoining $x_{i+1}, \ldots, x_n$, $R^{(i)}$ becomes equal to the determinant of the transition substitution from $\mathfrak{G}_{i-1}$ to $\mathfrak{M}_{i-1}$, and the degree of $R^{(i)}$ gives the number of residue classes, linearly independent over $P(u, x_{i+1}, \ldots, x_n)$, of $\mathfrak{G}_{i-1}$ modulo $\mathfrak{M}_{i-1}$, hence of $\mathfrak{g}_{i-1}$ modulo $\mathfrak{g}_i$; $R^{(i)}$ can be assumed integral and primitive in $x_{i+1}, \ldots, x_n$, and is then regular in x_i. $R^{(i)}$ can be computed as the greatest common divisor – in the polynomial sense – of all ϱ-rowed determinants of a module $\mathfrak{M}^*_{i-1}$, always existing, of rank ϱ, having the property that the residue-class system $\mathfrak{G}^*_{i-1}/\mathfrak{M}^*_{i-1}$ is isomorphic to $\mathfrak{G}_{i-1}/\mathfrak{M}_{i-1}$, where $\mathfrak{G}^*_{i-1}$ is understood as the fundamental module of $\mathfrak{M}^*_{i-1}$ (H.-N., Theorem VII and § 5).

6. Elementary-divisor form and norm (resultant form) of ideals. The product $E_\mathfrak{m} = E^{(1)}E^{(2)} \cdots E^{(n)}$ of all highest elementary divisors is called the elementary-divisor form, and the product $R_\mathfrak{m} = R^{(1)}R^{(2)} \cdots R^{(n)}$ of all individual norms the norm (resultant form)[6] of $\mathfrak{m}$. The elementary-divisor form and the norm are divisible by the ideal; the norm is divisible by the elementary-divisor form, and a power of the latter by the norm. More generally one still has:

$$\begin{aligned}
E^{(i)}\mathfrak{g}_{i-1} &\equiv 0 \pmod{\mathfrak{g}_i}, \qquad & E^{(n)} \cdots E^{(i)}\mathfrak{g}_{i-1} &\equiv 0 \pmod{\mathfrak{m}}, \\
R^{(i)}\mathfrak{g}_{i-1} &\equiv 0 \pmod{\mathfrak{g}_i}, \qquad & R^{(n)} \cdots R^{(i)}\mathfrak{g}_{i-1} &\equiv 0 \pmod{\mathfrak{m}},
\end{aligned} \tag{4}$$

[6] In H.-N. only the term resultant form is used; the term norm corresponds to the arithmetical properties.

(H.-N., Theorem VIII), and further: if $\mathfrak{m}$ is divisible by $\mathfrak{n}$, and the norms $R_\mathfrak{m}$ and $R_\mathfrak{n}$ agree, then the ideals $\mathfrak{m}$ and $\mathfrak{n}$ also agree (H.-N., Theorem IX). Taking into account that, for a divisor of $\mathfrak{m}$, agreement of $R^{(1)}, R^{(2)}, \ldots$ entails agreement of the fundamental ideals $\mathfrak{g}_1, \mathfrak{g}_2, \ldots$, and that always $\mathfrak{g}_0 = \mathfrak{o}$, one obtains correspondingly: if $\mathfrak{n}$ is a proper divisor of $\mathfrak{m}$, then the first individual norm of $\mathfrak{n}$ that differs from the corresponding one of $\mathfrak{m}$ becomes a proper divisor.[7] If $\mathfrak{m}$ contains a polynomial $G^{(i)} \neq 0$, then by definition $\mathfrak{g}_0, \mathfrak{g}_1, \ldots, \mathfrak{g}_{i-1}$ are equal to the unit ideal; consequently, by the property of individual norms stated in 5, namely to determine the number of linearly independent residue classes, $R^{(1)}, \ldots, R^{(i-1)}$ become equal to unity; hence also $E^{(1)}, \ldots, E^{(i-1)}$ become equal to unity.

7. Decomposition of the individual norms and elementary divisors; components of the fundamental ideals. By a prime function one understands a polynomial irreducible with respect to $P(u)$; by a primary function, the power of a prime function. A proper primary function is one that is not at the same time a prime function, where therefore powers higher than the first are involved. A decomposition of a polynomial into greatest primary factors is one in which every factor is a primary function, but the product of any two factors is no longer primary.[8] To the decomposition of the individual norm $R^{(i)} = N(\mathfrak{g}_{i-1} \mid \mathfrak{g}_i)$ into greatest primary factors $Q_\nu^{(i)}$ there corresponds a representation of $\mathfrak{g}_i$ as a least common multiple

$$\mathfrak{g}_i = [\mathfrak{r}_1, \mathfrak{r}_2, \ldots, \mathfrak{r}_s],$$

such that $Q_\nu^{(i)} \mathfrak{g}_{i-1} \equiv 0 \pmod{\mathfrak{r}_\nu}$; here $\mathfrak{r}_\nu$ possesses $\mathfrak{g}_{i-1}$ as its fundamental ideal of $(i-1)$-st stage, and agrees with its fundamental ideal of i-th stage; the $\mathfrak{r}_\nu$ are called the components of the fundamental ideals. The highest elementary divisor of $\mathfrak{r}_\nu$ becomes equal to the greatest common divisor – in the polynomial sense – of $Q_\nu^{(i)}$ and $E^{(i)}$, hence equal to a greatest primary factor of $E^{(i)}$. The $\mathfrak{r}_\nu$ are uniquely determined by their behavior, stated above, with respect to the fundamental ideals and by the requirement that either the individual norm or the highest elementary divisor should be a greatest primary factor of $R^{(i)}$ respectively $E^{(i)}$ (H.-N., Theorem X).

8. Concept of dimension. To the ideal $\mathfrak{m}$ the highest dimension $n-i$ is assigned if $R^{(i)}$ is the first individual norm different from unity – or, what by 6 is equivalent, if $\mathfrak{g}_i$ is the first fundamental ideal different from the unit ideal; the dimension -1 is assigned to the unit ideal $\mathfrak{o}$. If there is only one individual norm $R^{(i)}$ different from unity, so that $\mathfrak{g}_i = \mathfrak{g}_{i+1} = \cdots = \mathfrak{m}$, then the highest dimension is also called simply the dimension of $\mathfrak{m}$. If $\mathfrak{m}$ contains a polynomial $G^{(i)} \neq 0$, then it is of highest dimension at most $n-i$, as the remark at the end of 6 shows. If $\mathfrak{m}$ has highest dimension $n-i$ and $\mathfrak{n}$ is a divisor of $\mathfrak{m}$, then $\mathfrak{n}$ also has highest dimension at most $n-i$.

9. Norm (resultant form) and zeroes. By zeroes of an ideal $\mathfrak{m}$ one understands all systems of values of $x_1, \ldots, x_i$ belonging to a suitable algebraic extension field of $P(u; x_{i+1}, \ldots, x_n)$ $(i = 1, 2, \ldots, n)$ for which – after adjoining $x_{i+1}, \ldots, x_n$ to the coefficient domain – all polynomials from $\mathfrak{m}$ vanish. To each factor $R^{(i)}$ of the norm (resultant form), under this adjunction, there correspond as many zeroes of $\mathfrak{m}$ as the degree of $R^{(i)}$ indicates; and $R^{(i)}$, written in bilinear combinations $w_{\mu\nu}$ of the $u_{\mu\nu}$ and further indeterminates $v_{\mu\nu}$, admits the explicit decomposition

$$R^{(i)} = \prod_\nu (z_i - z_i^{(\nu)}) = \prod_\nu (v_{i1}x_1 + \cdots + v_{in}x_n - z_i^{(\nu)}), \tag{5}$$

where $x_1^{(\nu)}, \ldots, x_i^{(\nu)}$ denotes a system of associated zeroes (H.-N., Theorem XIII; a new foundation will be given in § 3). Thus among the zeroes of an ideal of highest dimension $n-i$ there are zeroes depending on $(n-i)$ parameters, but none depending on more parameters; this proves the agreement of the concept of dimension given in 8 with the usual formulation.

10. The ring domain of general ideal theory. The underlying domain is to be a commutative ring in which the theorem of the finite chain holds: every chain of ideals $\mathfrak{a}_1, \mathfrak{a}_2, \ldots, \mathfrak{a}_k, \ldots$, where each $\mathfrak{a}_i$ is a proper divisor – divisor in the ideal sense – of the immediately preceding one, breaks off after finitely many terms. This requirement is equivalent to the other, that every ideal possesses an ideal basis (Ideal Theory, Theorem I), and is therefore fulfilled in particular for the polynomial domain. In the following numbers 11, 12, 13, this general ring domain is assumed.

11. Prime ideals and primary ideals. An ideal $\mathfrak{p}$ is called a prime ideal if from the divisibility of a product by $\mathfrak{p}$ follows the divisibility of at least one factor; $\mathfrak{q}$ is called a primary ideal – or primary – if from

[7] On the other hand, later factors of $R_\mathfrak{n}$ can become multiples of the corresponding factors of $R_\mathfrak{m}$; for instance always when $\mathfrak{m}$ is a prime ideal and $\mathfrak{n}$ is not the unit ideal.

[8] The definition of prime function and primary function could also be formulated in exact analogy with 11 by replacing only the word ideal by polynomial. The greatest primary factors are the analogue of the greatest primary components of the decomposition in 12.

the divisibility of a product by $\mathfrak{q}$ follows the divisibility of one factor or of a power of every factor. In symbols:

$$a \not\equiv 0 \pmod{\mathfrak{p}}, \quad b \not\equiv 0 \pmod{\mathfrak{p}} \quad \text{implies} \quad ab \not\equiv 0 \pmod{\mathfrak{p}};$$

$$a \not\equiv 0 \pmod{\mathfrak{q}}, \quad b^x \not\equiv 0 \pmod{\mathfrak{q}} \text{ for every power } x \quad \text{implies} \quad ab \not\equiv 0 \pmod{\mathfrak{q}}.$$

To every primary ideal $\mathfrak{q}$ there is one and only one associated prime ideal $\mathfrak{p}$, which is a divisor of $\mathfrak{q}$ and a power of which is divisible by $\mathfrak{q}$, $\mathfrak{p} \neq \mathfrak{o}$, $\mathfrak{p}^\rho \equiv 0 \pmod{\mathfrak{q}}$; here $\mathfrak{p}$ consists of all ring elements of which a power is divisible by $\mathfrak{q}$. As the definition shows, every prime ideal is at the same time a primary ideal; proper primary ideals mean those that are not at the same time prime ideals, so that $\rho > 1$.

12. The decomposition theorem. A representation $\mathfrak{a} = [\mathfrak{q}_1, \ldots, \mathfrak{q}_s]$ of an ideal as a least common multiple is called shortest if no $\mathfrak{q}$ is contained in the least common multiple of the others – in its complement; it is called a representation by greatest primary components if every $\mathfrak{q}$ is primary, but the least common multiple of any two $\mathfrak{q}$'s is not primary. Every ideal admits a shortest representation by finitely many greatest primary components; for two different such representations, the number of components and the associated prime ideals, all of which are mutually distinct, agree. The prime ideals thus uniquely determined are to be called the prime ideals, or associated prime ideals, of the ideal (Ideal Theory, Theorem IX).

13. Isolated components of the decomposition. An ideal $\mathfrak{r} = [\mathfrak{q}_1, \ldots, \mathfrak{q}_i]$ is called an isolated component of the decomposition of $\mathfrak{a}$ if the $\mathfrak{q}_i$ occur in at least one shortest representation of $\mathfrak{a}$ by greatest primary components and satisfy the condition that none of their associated prime ideals is contained in one of the other prime ideals of $\mathfrak{a}$. The isolated components of the decomposition are uniquely determined by their prime ideals; in particular the isolated greatest primary components are uniquely determined (Ideal Theory, Theorem XIII).

14. Characteristic, perfect and imperfect fields, extension of the first kind. A field Ω is said to have characteristic zero if the prime field contained in Ω and derived from the unit is of the type of the field of rational numbers; it has characteristic p if this prime field is of the type of the residue-class system modulo a prime number p. A field Ω is called perfect if every prime function with coefficients from Ω decomposes in a suitable extension field into distinct linear factors; otherwise it is called imperfect. Every field of characteristic zero is perfect; a field of characteristic p is perfect if and only if, together with every element, it also contains its p-th root. A prime function in an imperfect field is an integral function of degree r in x^{p^f}, and decomposes in a suitable extension field into p^f-th powers of r distinct linear factors; f is called the exponent. A prime function is said to be of the first kind if it decomposes in a suitable extension field into distinct linear factors, so that the exponent f is zero; an extension is called of the first kind if every element is of the first kind, that is, is a zero of a prime function of the first kind. Every extension that arises by adjoining an element of the first kind is of the first kind; over perfect fields there are only extensions of the first kind (Steinitz, § 11 and § 13).

15. Reduced degree and exponent of a finite extension. If $\mathcal{K}$ is a finite extension of an imperfect field Ω, then $\mathcal{K}$ contains a subfield $\mathcal{K}_0$ of degree r that is of the first kind with respect to Ω and consists of all and only the elements of the first kind in $\mathcal{K}$. The p^f-th power of every element of $\mathcal{K}$ belongs to $\mathcal{K}_0$, and there are elements in $\mathcal{K}$ that are exactly of degree rp^f. r is called the reduced degree, f the exponent of the finite extension (Steinitz, § 14, 1 and § 14, 5).

§ 2. Elementary-Divisor Form and Norm of the Prime Ideals and Primary Ideals. Divisors of the Prime Ideals

From now on we shall throughout be concerned with the polynomial domain defined in § 1, 1. First it is to be shown that, also for prime ideals and primary ideals, one may restrict oneself to transformed ideals, according to

Lemma I. The transformed ideal $\mathfrak{p}$ of a prime ideal $\bar{\mathfrak{p}}$ in $\mathfrak{S}$ becomes a prime ideal; the transformed $\mathfrak{q}$ of a primary ideal $\bar{\mathfrak{q}}$ in $\mathfrak{S}$ becomes a primary ideal. In other words, the property of being a prime ideal or a primary ideal is preserved when the $u_{\mu\nu}$ are adjoined to P.

Indeed suppose that $\mathfrak{a} \not\equiv 0 \pmod{\mathfrak{p}}$, $\mathfrak{b} \not\equiv 0 \pmod{\mathfrak{p}}$, where $\mathfrak{a}$ and $\mathfrak{b}$ need not be transformed ideals; say $a(x) \not\equiv 0 \pmod{\mathfrak{p}}$ and $b(x) \not\equiv 0 \pmod{\mathfrak{p}}$, with $a(x)$ and $b(x)$ polynomials from $\mathfrak{a}$ and $\mathfrak{b}$. By inversion of (1) according to (1') one obtains – with T understood, without loss of generality, as power products of the $t_{\mu\nu}$ –

$$a(x) = \sum T_i a_i(y), \qquad b(x) = \sum T'_i b_i(y),$$

and at least one $a_i(y)$ and one $b_i(y)$ must fail to be divisible by $\bar{\mathfrak{p}}$. Let, for example, $a_h(y) \not\equiv 0 \pmod{\bar{\mathfrak{p}}}$ and $b_k(y) \not\equiv 0 \pmod{\bar{\mathfrak{p}}}$ be the respective highest terms under some lexicographic ordering of the T's; then also

$a_h(y)b_k(y) \not\equiv 0 \pmod{\bar{\mathfrak{p}}}$, and consequently $a(x)b(x) \not\equiv 0 \pmod{\mathfrak{p}}$ and hence $\mathfrak{ab} \not\equiv 0 \pmod{\mathfrak{p}}$, proving that $\mathfrak{p}$ is prime. The proof evidently rests on the fact that the residue-class system modulo a prime ideal forms a ring without zero divisors, a property preserved by adjoining indeterminates.

Correspondingly suppose that $\mathfrak{a} \not\equiv 0 \pmod{\mathfrak{q}}$ and $\mathfrak{b}^x \not\equiv 0 \pmod{\mathfrak{q}}$ for every power x; then from the existence of the ideal basis it follows that there must exist an $a(x)$ from $\mathfrak{a}$ and a $b(x)$ from $\mathfrak{b}$ such that $a(x) \not\equiv 0 \pmod{\mathfrak{q}}$ and $b(x)^x \not\equiv 0 \pmod{\mathfrak{q}}$ for every power x; for under the contrary assumption, a power of $\mathfrak{b}$ would be divisible by $\mathfrak{q}$.

Putting $a(x)b(x) = c(x)$, let $\mathfrak{a}^*, \mathfrak{b}^*, \mathfrak{c}^*$ denote the ideals respectively derived from the coefficients $a_i(y), b_j(y), c_k(y)$ of $a(x), b(x), c(x)$. Then $\mathfrak{b}^{*x} \not\equiv 0 \pmod{\bar{\mathfrak{q}}}$ for every power x, since otherwise $b(x)^x$ would be divisible by $\mathfrak{q}$; because $\mathfrak{a}^* \not\equiv 0 \pmod{\bar{\mathfrak{q}}}$, one must also have $\mathfrak{a}^*\mathfrak{b}^{*\lambda} \not\equiv 0 \pmod{\bar{\mathfrak{q}}}$ for every power λ. But by the Dedekind-Mertens theorem[9] there exists an exponent λ such that $\mathfrak{a}^*\mathfrak{b}^{*\lambda} \equiv \mathfrak{c}^*\mathfrak{b}^{*\lambda-1}$; hence $\mathfrak{c}^* \not\equiv 0 \pmod{\bar{\mathfrak{q}}}$. This gives $c(x) \not\equiv 0 \pmod{\mathfrak{q}}$ and consequently $\mathfrak{ab} \not\equiv 0 \pmod{\mathfrak{q}}$, proving that $\mathfrak{q}$ is primary.

Also in representing an ideal as a least common multiple one may restrict oneself to transformed ideals, according to

Lemma II. From a representation $\overline{\mathfrak{m}} = [\mathfrak{c}_1, \ldots, \mathfrak{c}_s]$ in $\mathfrak{S}$ follows a representation $\mathfrak{m} = [\mathfrak{c}'_1, \ldots, \mathfrak{c}'_s]$ in $\mathfrak{S}'$, where $\mathfrak{c}'_i$ denotes the transformed ideal of $\mathfrak{c}_i$. In particular, therefore, in every shortest representation by greatest primary components these may be assumed transformed.

Indeed $\mathfrak{m}$ is divisible by $[\mathfrak{c}'_1, \ldots, \mathfrak{c}'_s]$, since every polynomial $f(x)$ from $\mathfrak{m}$ – after multiplication, if necessary, by a suitable power of $U(u)$ – is of the form $\sum U_i(u)f_i(y)$, where each $f_i(y)$, as a polynomial from $\overline{\mathfrak{m}}$, is by assumption divisible by all the $\mathfrak{c}_i$. Conversely, if $f(x)$ is divisible by every $\mathfrak{c}'_i$, then it has the above form and is therefore divisible by $\mathfrak{m}$. Thus if one starts from a decomposition into greatest primary components in $\mathfrak{S}$, one obtains a decomposition $\mathfrak{m} = [\mathfrak{q}'_1, \ldots, \mathfrak{q}'_s]$ in $\mathfrak{S}'$; here the $\mathfrak{q}'_i$, as transformed ideals of primary ideals, are primary by Lemma I, and they are greatest primary components, since distinct associated prime ideals $\bar{\mathfrak{p}}$ in $\mathfrak{S}$ correspond to distinct $\mathfrak{p}$ in $\mathfrak{S}'$.

We now need a first characterization of prime ideals and primary ideals by elementary-divisor form and norm, still without complete separation of prime and primary. In exact analogy with ideal theory in algebraic number fields,[10] one has

Theorem I. The elementary-divisor form of a prime ideal becomes equal to a prime function, and the norm to a power of this prime function. For primary ideals, elementary-divisor form and norm become primary functions, namely powers of the same prime function, which itself is the elementary-divisor form of the associated prime ideal. For prime ideals and primary ideals, highest dimension coincides with dimension simpliciter; this dimension agrees for a primary ideal and its associated prime ideal.

Theorem I is evidently fulfilled for the unit ideal, whose elementary-divisor form and norm become unity. Now let the prime ideal $\mathfrak{p}$ have highest dimension $(n-i)$; by (4), § 1, 6, one obtains

$$E^{(i)}\mathfrak{g}_{i-1} \not\equiv 0 \pmod{\mathfrak{p}}, \quad \text{but} \quad E^{(i)}\mathfrak{g}_i \equiv 0 \pmod{\mathfrak{p}},$$

since otherwise $\mathfrak{p}$ would, by § 1, 8, have highest dimension at most $n-i-1$. Hence $\mathfrak{g}_i \equiv 0 \pmod{\mathfrak{p}}$, so that $\mathfrak{p}$ is identical with its fundamental ideal of i-th stage and consequently also with $\mathfrak{g}_{i+1}, \mathfrak{g}_{i+2}, \ldots$. Thus $R^{(i+1)}, \ldots, R^{(n)}$ become equal to unity, and hence also $E^{(i+1)}, E^{(i+2)}, \ldots$, as divisors of $R^{(i+1)}, R^{(i+2)}, \ldots$; therefore the elementary-divisor form becomes equal to $E^{(i)}$, which must be equal to a prime function $P^{(i)}$. For by § 1, 5 and taking into account that $\mathfrak{g}_{i-1}$ becomes the unit ideal, $E^{(i)}$ is defined as the greatest common divisor – in the polynomial sense – of all $B^{(i)}$ divisible by $\mathfrak{p}$; from $E^{(i)} = E_1^{(i)}E_2^{(i)} \equiv 0 \pmod{\mathfrak{p}}$ it would follow that $E^{(i)}$ must be contained in one of its factors $E_1^{(i)}$ or $E_2^{(i)}$. Since, however, a power of $E^{(i)}$ is divisible by $R^{(i)}$, $R^{(i)}$ becomes a power – possibly the first power – of this prime function $P^{(i)}$; at the same time $R^{(i)}$ becomes the norm of $\mathfrak{p}$. Since only one individual norm different from unity occurs, the highest dimension of $\mathfrak{p}$ finally agrees with its dimension simpliciter.

Correspondingly, let the primary ideal $\mathfrak{q}$ have highest dimension $n-i$; then, as above,

$$(E^{(i)}\mathfrak{g}_{i-1})^x \not\equiv 0 \pmod{\mathfrak{q}}, \quad \text{but} \quad (E^{(i)}\mathfrak{g}_i)^x \equiv 0 \pmod{\mathfrak{q}}$$

for every power x; and consequently, as above, $\mathfrak{q} = \mathfrak{g}_i$, its elementary-divisor form is $E^{(i)}$, its norm is $R^{(i)}$, and its highest dimension is its dimension simpliciter. This dimension agrees with that of the associated

[9] Compare H.-N., p. 63 and note 10).

[10] As the highest elementary divisor of ideals in algebraic number fields one has to regard the least integral rational number divisible by the ideal, hence in particular the prime number divisible by a prime ideal or the prime-power divisible by a primary ideal (a power of a prime ideal). In fact every ideal $\mathfrak{a}^*$ is also a module A^* of linear forms in the elements $\omega_1, \ldots, \omega_n$ of a field basis, with $(\omega_1, \ldots, \omega_n)$ the fundamental module of A^*; the least integral rational number divisible by $\mathfrak{a}^*$ therefore becomes the highest elementary divisor of this module A^*, whereas the norm of $\mathfrak{a}^*$ becomes the product of all elementary divisors of A^*.

prime ideal; for from $\mathfrak{p}^\rho \equiv 0 \pmod{\mathfrak{q}}$, $\mathfrak{q} \equiv 0 \pmod{\mathfrak{p}}$ it follows that the dimension $(n-i)$ of $\mathfrak{p}$ is at most equal to that of $\mathfrak{q}$, and that of $\mathfrak{q}$ at most equal to that of $\mathfrak{p}^\rho$; from $(P^{(i)})^\rho \equiv 0 \pmod{\mathfrak{p}^\rho}$, where $P^{(i)}$ denotes the elementary-divisor form of $\mathfrak{p}$, the latter highest dimension is found to be at most $n-i$, and thus $n-i$ is the dimension of $\mathfrak{p}$ and $\mathfrak{q}$. From $(P^{(i)})^\rho \equiv 0 \pmod{\mathfrak{q}}$ it follows further that $E^{(i)}$ – as the greatest common divisor of all $B^{(i)}$ from $\mathfrak{q}$ – becomes a power of $P^{(i)}$, which may also be the first power, and that the same holds for $R^{(i)}$. This proves Theorem I in all its parts.

A characterization of prime ideals by means of the concept of dimension is given by

Theorem II. A prime ideal has no proper divisor of the same highest dimension; conversely, every ideal with this property is prime.

The proof rests on

Lemma III. If a prime ideal $\mathfrak{p}$ of polynomials with coefficients from a field Ω has only finitely many residue classes linearly independent over Ω, then $\mathfrak{p}$ has no proper divisor different from the unit ideal; in other words, the residue-class system modulo $\mathfrak{p}$ forms a field.

The hypothesis says that there is a finite number k such that among any $(k+1)$ residue classes there is a linear dependence with coefficients from Ω – more precisely, with coefficients from the residue-class field (Ω) represented by all elements of Ω – but that at least one system of k residue classes linearly independent over Ω exists. It follows immediately that all residue classes can be expressed linearly by any chosen system of k linearly independent ones. Now let $\mathfrak{a}$ be a proper divisor of $\mathfrak{p}$, and let $a(z) \not\equiv 0 \pmod{\mathfrak{p}}$ be a polynomial from $\mathfrak{a}$; from $c_1 f_1(z) + \cdots + c_k f_k(z) \equiv 0 \pmod{\mathfrak{p}}$ it follows that

$$c_1 a(z) f_1(z) + \cdots + c_k a(z) f_k(z) \equiv 0 \pmod{\mathfrak{p}}.$$

That means that, together with the residue classes of $f_1, \ldots, f_k$, the residue classes of $af_1, \ldots, af_k$ also form a system of k linearly independent ones, through which in particular the unit class can be linearly represented. Hence $\mathfrak{a}$ is the unit ideal; expressed differently, the residue-class system forms a field, since for $a(z) \not\equiv 0 \pmod{\mathfrak{p}}$ there is always an $f(z)$ such that $1 \equiv a(z)f(z) \pmod{\mathfrak{p}}$.

For the proof of the first part of Theorem II it remains only to observe that every prime ideal before dimension $(n-i)$ – more generally every ideal of highest dimension $(n-i)$ – has only finitely many residue classes linearly independent over $P(u; x_{i+1}, \ldots, x_n)$, namely, by § 1, 5, as many as the degree of $R^{(i)}$ indicates, since here $\mathfrak{g}_{i-1}$ becomes the unit ideal. Every proper divisor $\mathfrak{a}$ of $\mathfrak{p}$ therefore becomes the unit ideal when $x_{i+1}, \ldots, x_n$ are adjoined to $P(u)$; expressed differently, the fundamental ideal of i-th stage of $\mathfrak{a}$ becomes the unit ideal, which means that the highest dimension of $\mathfrak{a}$ is at most $n-i-1$. In order that the assertion also hold for a non-transformed ideal $\mathfrak{a}$ as divisor, one has only to pass, according to § 1, 2, to a new transformed domain.

Conversely suppose now that $\mathfrak{p}$ has no proper divisor of the same highest dimension $n-i$, and let $\mathfrak{a} \not\equiv 0 \pmod{\mathfrak{p}}$ and $\mathfrak{b} \not\equiv 0 \pmod{\mathfrak{p}}$, with $\mathfrak{a}$ and $\mathfrak{b}$ again assumed to be transformed ideals. Then, by hypothesis, since $(\mathfrak{a}, \mathfrak{p})$ and $(\mathfrak{b}, \mathfrak{p})$, as greatest common divisors – in the ideal sense – become proper divisors of $\mathfrak{p}$, one has

$$E^{(n)} \cdots E^{(i+1)} \mathfrak{g}_i \equiv 0 \pmod{(\mathfrak{a}, \mathfrak{p})}, \qquad E^{(n)} \cdots E^{(i+1)} \mathfrak{g}_i \equiv 0 \pmod{(\mathfrak{b}, \mathfrak{p})}.$$

This gives

$$E^{(n)} \cdots E^{(i+1)} \mathfrak{g}_i \equiv 0 \pmod{(\mathfrak{ab}, \mathfrak{p})},$$

and consequently necessarily $\mathfrak{ab} \not\equiv 0 \pmod{\mathfrak{p}}$, since otherwise $\mathfrak{p}$ would have smaller highest dimension than $n-i$. Since $\mathfrak{a}, \mathfrak{b}$ are transformed ideals, $\mathfrak{p}$, and by Lemma I also $\bar{\mathfrak{p}}$, are prime ideals. This proves Theorem II.

§ 3. Zeroes of the Prime Ideals and Primary Ideals

The passage to a direct foundation of the theory of zeroes (§ 1, 9), at first for prime ideals, is formed by the following lemma, which extends Lemma III and is valid for prime ideals in arbitrary rings.

Lemma IV. The residue-class system modulo every prime ideal different from the unit ideal can be extended by quotient formation to a field, the residue-class field of the prime ideal.

The residue-class system modulo an arbitrary ideal forms a ring, since sum, difference and product again belong to the system, and the associative, commutative and distributive laws are preserved on passage to residue classes. If the ideal is specifically prime, this ring has no zero divisors; for the definition of prime ideals (§ 1, 11) says that the product of non-vanishing residue classes likewise does not vanish. If the prime ideal is different from the unit ideal, then this ring contains at least one element different from the zero element, and can therefore be extended to a field by quotient formation – adjunction of pairs of elements;[11] thus the residue-class field is defined.

We first fix the notation that will be kept throughout what follows.

Notation. Residue classes shall generally be denoted by putting any representative of the class in parentheses, $(x), \ldots, (a(x)), \ldots$; similarly, fields of residue classes shall be denoted by parentheses. In particular, (P), $(P(u))$, $(P(u, x_{i+1}, \ldots, x_n))$ denote the residue-class fields consisting of all and only those residue classes that can be represented by elements from P, $P(u)$, $P(u, x_{i+1}, \ldots, x_n)$.

We also recall the following. A field $\mathcal{R}$ is said to have algebraic rank (transcendence degree) k with respect to a base field Ω if $\mathcal{R}$ contains at least one system of k quantities algebraically independent over Ω – transcendental quantities – but every $k+1$ quantities from $\mathcal{R}$ are algebraically dependent over Ω. If $\mathcal{R}$ can be represented as an algebraic extension field of a subfield $\Omega(z_1, \ldots, z_s)$, where the z are algebraically independent – transcendental – over Ω, then necessarily $s = k$;[12] likewise the algebraic rank of $\mathcal{R}(u)$ over $\Omega(u)$ is equal to k when u denotes indeterminates not contained in $\mathcal{R}$.

The construction of the field of zeroes now proceeds in complete analogy with the usual route for prime functions of one indeterminate,[13] according to

Theorem III. To the residue-class field $(\mathcal{R})$ of a prime ideal $\mathfrak{p}$ of dimension $n-i$ there can be assigned, isomorphically, a field of zeroes R of $\mathfrak{p}$, which has algebraic rank $n-i$ – depends on $n-i$ parameters – and in particular can be regarded as a finite extension field of $P(u, x_{i+1}, \ldots, x_n)$. The isomorphism between $(\mathcal{R})$ and R includes that between (P) and P, $(P(u))$ and $P(u)$, and, when $x_{i+1}, \ldots, x_n$ are taken as parameters, also that between $(P(u, x_{i+1}, \ldots, x_n))$ and $P(u, x_{i+1}, \ldots, x_n)$. Conversely, every field of zeroes of algebraic rank $n-i$ – depending on $n-i$ parameters – is isomorphic to the residue-class field $(\mathcal{R})$.

Theorem III yields as a corollary the familiar theorem: if an ideal $\mathfrak{a}$ vanishes in all zeroes of a prime ideal $\mathfrak{p}$, then $\mathfrak{a}$ is divisible by $\mathfrak{p}$.

For the proof, first note that the residue-class field $(\mathcal{R})$ has algebraic rank $n-i$ with respect to $(P(u))$. Indeed, the classes $(x_{i+1}), \ldots, (x_n)$ are algebraically independent with respect to $(P(u))$, since $\mathfrak{p}$, being of dimension $n-i$, contains no polynomial free of $x_1, \ldots, x_i$; every further class, however, is dependent on these. For from the membership of the elementary-divisor form $P^{(i)}$ in $\mathfrak{p}$ it follows, by § 1, 2, also for $\lambda = 1, 2, \ldots, i$ that polynomials belong to $\mathfrak{p}$ which depend respectively on $x_\lambda, x_{i+1}, \ldots, x_n$. Thus $(\mathcal{R})$ becomes a finite extension field of $(P(u, x_{i+1}, \ldots, x_n))$; by § 1, 5 – taking into account that $\mathfrak{g}_{i-1}$ is equal to the unit ideal and $\mathfrak{g}_i$, by Theorem I, to $\mathfrak{p}$ – the degree over this subfield is equal to the degree of the norm.

To pass to an isomorphic field of zeroes, one further observes that $P(u, x_{i+1}, \ldots, x_n)$ and $(P(u, x_{i+1}, \ldots, x_n))$ are related isomorphically by assigning to each element of $P(u, x_{i+1}, \ldots, x_n)$ the residue class represented by that element. First, under this assignment, the integral domains consisting of polynomials in $u, x_{i+1}, \ldots, x_n$ and respectively in $(u), (x_{i+1}), \ldots, (x_n)$ correspond isomorphically, since $\mathfrak{p}$ contains no polynomial free of $x_1, \ldots, x_i$, and therefore distinct polynomials from $P(u, x_{i+1}, \ldots, x_n)$ correspond to distinct classes; from isomorphic integral domains, however, isomorphic fields arise by quotient formation. The isomorphic relation between $P(u, x_{i+1}, \ldots, x_n)$ and $(P(u, x_{i+1}, \ldots, x_n))$ includes that between P and (P), $P(u)$ and $(P(u))$. To extend this isomorphism to one between $(\mathcal{R})$ and a field of zeroes R, let certain new elements $\xi_1, \ldots, \xi_i$ be assigned isomorphically to the classes $(x_1), \ldots, (x_i)$; that is, to each polynomial $(a(x))$ there corresponds a polynomial $a(\xi_1, \ldots, \xi_i, x_{i+1}, \ldots, x_n)$, and sums and products correspond to sums and products. By quotient formation – where one may restrict oneself to denominators from the subfield $(P(u, x_{i+1}, \ldots, x_n))$ respectively $P(u, x_{i+1}, \ldots, x_n)$ – to the residue-class field $(\mathcal{R})$ there then corresponds a field R, a finite extension of $P(u, x_{i+1}, \ldots, x_n)$ of the degree of the norm; it can be called the field of zeroes. For the assignment says that $f(\xi_1, \ldots, \xi_i, x_{i+1}, \ldots, x_n)$ vanishes as soon as $f(x) \equiv 0 \pmod{\mathfrak{p}}$. In place of $(P(u, x_{i+1}, \ldots, x_n))$, any other subfield of $(\mathcal{R})$ arising by adjoining $n-i$ algebraically independent quantities could occur throughout the argument.

Conversely it is easy to show that every field of zeroes R of algebraic rank $n-i$ is isomorphic to the residue-class field. Since R can be derived rationally, with coefficients from $P(u)$, from the quantities $\xi_1, \ldots, \xi_i$, among these quantities there must be $n-i$ algebraically independent, hence transcendental, elements with respect to $P(u)$. In particular this must hold for $\xi_{i+1}, \ldots, \xi_n$, since, because $P^{(i)} \equiv 0 \pmod{\mathfrak{p}}$, it follows as above that R is an algebraic extension field of $P(u, \xi_{i+1}, \ldots, \xi_n)$. Since by hypothesis $f(\xi)$ vanishes for every

[11] Steinitz, § 3. The assumption of a non-zero element in the original integral domain suffices, since then the existence of the unit is secured by quotient formation, and hence the original domain becomes a subdomain of the field; Steinitz assumes the existence of the unit in the integral domain.

[12] For a proof of this familiar fact valid in arbitrary characteristic, compare Steinitz, § 22, 9. – On adjoining the u, the algebraic rank cannot increase, since only rational combinations of the original elements with coefficients from $\Omega(u)$ are involved; nor can it decrease, since every relation between elements of $\mathcal{R}$ with coefficients from $\Omega(u)$ decomposes into finitely many relations with coefficients from Ω.

[13] Steinitz, § 6.

polynomial $f(x)$ in $\mathfrak{p}$, to every class of the polynomial domain of the (x) contained in $(\mathcal{R})$ there corresponds one and only one element of R which is a polynomial in the ξ; and sums and products correspond to sums and products. But different classes correspond to different elements of R; otherwise a non-zero class, and hence after suitable multiplication also a class represented by a polynomial from $(P(u, x_{i+1}, \ldots, x_n))$, would correspond to the element zero in R, and contrary to the hypothesis the quantities $\xi_{i+1}, \ldots, \xi_n$ would satisfy an algebraic dependence. This assignment exhausts all polynomials in the ξ contained in R; by quotient formation – where one may again restrict oneself to denominators from $(P(u, x_{i+1}, \ldots, x_n))$ respectively $P(u, x_{i+1}, \ldots, x_n)$ – isomorphic fields arise from these isomorphic integral domains.

Now let the ideal $\mathfrak{a}$ vanish in all zeroes of $\mathfrak{p}$, in particular also in all zeroes depending on $n - i$ parameters. If $a(x)$ is any polynomial from $\mathfrak{a}$, then $a(\xi)$ vanishes; because of the isomorphism between R and $(\mathcal{R})$, the class $(a(x))$ is thus the zero class, and $a(x)$, hence $\mathfrak{a}$, is divisible by $\mathfrak{p}$. This proves Theorem III and its corollary.

To pass from Theorem III to the decomposition given in § 1, 9, first for prime ideals, and in order to be able to make more precise statements about the exponents, a further investigation of the elementary-divisor form is needed, according to

Lemma V. If in characteristic p the elementary-divisor form $E^{(i)}$ of a prime ideal or primary ideal – more generally, of an ideal $\mathfrak{a}$ for which highest dimension and dimension simpliciter coincide – has exponent f with respect to x_i, hence is an integral function of $x_i^{p^f}$, then it also has exponent f with respect to $x_{i+1}, \ldots, x_n$ and with respect to the indeterminates $u_{\mu\nu}$ respectively $t_{\mu\nu}$ occurring in $E^{(i)}$. In particular, if the coefficient domain is a perfect field, the elementary-divisor form $P^{(i)}$ of a prime ideal always has exponent zero with respect to x_i, hence is a prime function of the first kind.

For this it is to be observed that $P(u, x_{i+1}, \ldots, x_n)$ becomes imperfect as soon as P is a perfect field of characteristic p, so that it does not follow directly from § 1, 14 that $P^{(i)}$ is of the first kind. Let $E^{(i)}$ have exponent f with respect to x_i, but let it contain another indeterminate x_{i+1} with exponent f'. Then – by interchanging the $u_{\mu\nu}$ respectively $t_{\mu\nu}$ – there also exists, by § 1, 2, a non-zero polynomial $G^{(i)}$, divisible by the ideal, that is an integral function of $x_{i+1}^{p^{f'}}$, and which, by definition of the elementary-divisor form, is divisible by $E^{(i)}$. But since $G^{(i)}$ has the same degree as $E^{(i)}$, it must agree with $E^{(i)}$ up to a factor from $P(u)$; hence necessarily $f' = f$. Thus $E^{(i)}$ – which may be assumed integral and primitive in the $t_{\mu\nu}$ – is of the form

$$E^{(i)} = \sum c_\rho(t)\, x_i^{\alpha_\rho p^f}\, x_{i+1}^{\beta_\rho p^f} \cdots x_n^{\gamma_\rho p^f} = \sum c_\rho(t)\, X_\rho^{p^f},$$

where the X_ρ are power products of the x, and the X_ρ are divisible by $\mathfrak{a}$. Therefore one obtains a polynomial again belonging to $\mathfrak{a}$ if in the X_ρ every exponent of the individual t is replaced by the greatest multiple of p^f contained in it; as the displayed representation shows, this amounts to replacing, in $c_\rho(t)$, every exponent of the t by the greatest multiple of p^f contained in it. By this substitution $E^{(i)}$ passes into a polynomial in $x_i^{p^f}, \ldots, x_n^{p^f}$ which, by definition, is divisible by $E^{(i)}$, and, because of the assumed primitivity, also with respect to the t; hence it must be identical with $E^{(i)}$, since the degree has increased in no argument. In particular, if P is perfect, then $E^{(i)}$, as an integral function of $x_i^{p^f}$, is a p^f-th power; hence, if a prime ideal is involved, necessarily $f = 0$, and the elementary-divisor form $P^{(i)}$ becomes a prime function of the first kind.

The decomposition is now given by

Theorem IV. The elementary-divisor form $P^{(i)}$ of a prime ideal admits, in the Galois field derived from the field of zeroes, the explicit decomposition

$$P^{(i)} = \prod_\nu (x_i - t_{i+1}x_{i+1}^{(\nu)} - \cdots - t_n x_n^{(\nu)})^e = \prod_\nu (t_i(y_i - y_i^{(\nu)}) + \cdots + t_n(y_n - y_n^{(\nu)}))^e,$$

where $y_1^{(\nu)}, \ldots, y_n^{(\nu)}$ denotes an associated system of zeroes of the original indeterminates y, independent of $t_i, \ldots, t_n$; distinct ν correspond to distinct linear factors; and the exponent e has the value one when P is perfect and the value p^f $(f > 0)$ when it is imperfect. Thereby, after adjoining new indeterminates v, the decomposition of the elementary-divisor form written in bilinear combinations of the u and v in place of the $u_{\mu\nu}$ is also determined:

$$P^{(i)}(w) = \prod_\nu (z_i - z_i^{(\nu)})^e = \prod_\nu (v_{i1}(y_1 - y_1^{(\nu)}) + \cdots + v_{in}(y_n - y_n^{(\nu)}))^e,$$

where e has the same meaning as above. Likewise the decomposition of the norm of a prime ideal, or of a primary ideal with $\mathfrak{p}$ as associated prime ideal – as a power of $P^{(i)}$ – is also given.

As a corollary of Theorem IV one also obtains: if two prime ideals agree in their elementary-divisor form $P^{(i)}$, then they are identical.

The proof of Theorem IV will amount to proving that the elementary-divisor form $P^{(i)}$ is identical with the fundamental equation of the extension field $(P(u, x_{i+1}, \ldots, x_n), (x_i))$ over $(P(u, x_{i+1}, \ldots, x_n))$. First, to gain insight into the dependence on the indeterminates $u_{\mu\nu}$ respectively $t_{\mu\nu}$, pass to the residue-class field $(\mathcal{S})$ of $\mathfrak{p}$, which, by what was said in the definition of algebraic rank, must have algebraic rank $n-i$ with respect to (P), where (P) denotes the subfield represented by elements of P. For $(\mathcal{S})$ arises by adjoining the indeterminates u respectively t to a subfield isomorphic to $(\mathcal{R})$, obtained by quotient formation from all and only those classes representable by polynomials from $\mathfrak{S}$, with (P) and (P) corresponding to one another. The algebraic rank of $(\mathcal{S})$ with respect to $(P(u))$ is therefore equal to that of this subfield with respect to (P), and hence, by the isomorphism, to that of $(\mathcal{R})$ with respect to (P). Since $(\mathcal{S})$ can be derived rationally, with coefficients from (P), from the classes $(y_1), \ldots, (y_n)$, there must be among these classes $n-i$ algebraically independent ones, and $(\mathcal{S})$ arises as a finite algebraic extension field of the subfield generated by adjoining these $n-i$ classes to (P).[14]

If one adjoins only the indeterminates $t_{\mu\nu}$ occurring in $x_{i+1}, \ldots, x_n$, then again a residue-class field arises which contains the classes $(y_1), \ldots, (y_n)$ and $(x_{i+1}), \ldots, (x_n)$ and becomes a finite algebraic extension field of $(P(t_{\mu\nu}, x_{i+1}, \ldots, x_n))$; these are fields isomorphic to subfields of $(\mathcal{S})$, not identical with them, since for different indeterminates the residue classes are represented by the same elements but are not identical, because they do not contain the same elements. Thus in the dependence of the classes (y) on $(P(t_{\mu\nu}, x_{i+1}, \ldots, x_n))$ only the indeterminates $t_{\mu\nu}$ enter. Now form, by adjoining $t_{i1}, \ldots, t_{in}$, the class $(x_i) = (t_{i1}y_1 + \cdots + t_{in}y_n)$; let $(x_{i\nu}) = (t_{i1}y_{1\nu} + \cdots + t_{in}y_{n\nu})$ be the r distinct conjugate values lying in the Galois field with respect to $(P(t_{\mu\nu}, x_{i+1}, \ldots, x_n))$. Then – according as one is dealing with extensions of the first kind or not – the product over the factors $x_i - (x_{i\nu})$, or over their p^f-th powers, is a prime function (G) with respect to $(P(t_{\mu\nu}, x_{i+1}, \ldots, x_n))$; for, since there are elements of degree $r \cdot p^f$ (§ 1, 15), (x_i) must necessarily be one such element, since every element arises by specializing the $t_{\mu\nu}$; but (x_i) is a zero of (G). Passing to the isomorphic field of zeroes of $\mathfrak{p}$ and to the Galois field derived from it, G therefore admits a decomposition into linear factors $x_i - t_{i1}y_{1\nu} - \cdots - t_{in}y_{n\nu}$ respectively their p^f-th powers. Now, since (G) vanishes for $x_i = (x_i)$, $G(x_i)$ is divisible by $\mathfrak{p}$ and hence by $P^{(i)}$; and, as a prime function with respect to $P(t_{\mu\nu}, x_{i+1}, \ldots, x_n)$ and as primitive in x_i, G must become identical with $P^{(i)}$. Thus the first decomposition of Theorem IV is proved.

To pass to the second decomposition, note that the property of being, or not being, an extension of the first kind is preserved when indeterminates are adjoined. Therefore, after adjoining $v_1, \ldots, v_n$ and all the $t_{\mu\nu}$, form the class $(z) = (v_1x_1 + \cdots + v_ix_i)$ and the conjugate classes $(z_\nu) = (v_1x_{1\nu} + \cdots + v_ix_{i\nu})$. Then the product over all $z - (z_\nu)$, or over their p^f-th powers, is again a prime function (H) that vanishes for $z = (z)$. Hence H is representable as a product of linear factors, as above, and is identical with the elementary-divisor form of the prime ideal derived from $\mathfrak{p}$ by composing the transformation (1) with $z = v_1x_1 + \cdots + v_nx_n$; this elementary-divisor form, however – because of the mutual specialization – is obtained from $P^{(i)}$ by replacing the $u_{\mu\nu}$ by certain bilinear combinations of the u and v.[15] With these decompositions, the decomposition of the norm as a power of $P^{(i)}$ is also given,[16] and likewise that of the norm and elementary-divisor form of a primary ideal with $\mathfrak{p}$ as associated prime ideal.

Finally, if two prime ideals agree in their elementary-divisor form $P^{(i)}$, then, as a consequence of the above decomposition, they agree in their zeroes; hence they are mutually divisible by one another and are therefore identical.

[14] This remark shows that Theorem III with its corollary could have been stated and proved directly for the field of zeroes R of $\mathfrak{p}$. Only by passing to the transformed ideal is it achieved that in the greatest primary components of equal dimension of an arbitrary ideal the same parameters occur among the zeroes; and only thereby does the connection with norm and elementary-divisor form of an arbitrary ideal result.

[15] H.-N., § 7.

[16] For extensions of the first kind, hence in particular over perfect fields, the first power is involved; over imperfect fields the p^f-th power is involved, as will be shown in § 6, Theorems XIII and XIV.